Weapon Threat Detection: From Sandbox to Production

Executive Summary

This report documents the engineering journey behind a real-time Weapon Threat Detection system built for CCTV surveillance. The pipeline is built on a YOLO11s backbone, transfer-learned from COCO-pretrained weights, and developed through a deliberately staged process: architecture search and hyperparameter tuning on a small, statistically representative sample first, followed by validated scale-up to the full production dataset. This document narrates that journey — from the 20,000-image sandbox, through the engineering trade-offs required to scale to 130,000+ images, to the current status of the production run — and lays out the road ahead.

Headline result from the sandbox phase: the corrected sample-phase model (IMPROVED_1 / EXP-IMPROVED-09) beat the original baseline on every core metric — precision +10.2 points, recall +12.3 points, mAP50 +15.3 points, mAP50-95 +13.4 points — clearing the Stage 08 validation gate (mAP50-95 ≥ 0.315) with a 42.6% margin (0.4491 vs. 0.315) and unlocking full-dataset training.

Chapter 1: The Sandbox — 20K Sample Experiments

Before touching the full corpus, the team built a 20,000-image sample — roughly 18.3% of the 109,403-image cleaned dataset — using deterministic, diversity-aware stratified sampling (never plain random sampling). The sample was validated at 99.59% overall representativeness, with class balance and object-size distribution both above 99%; the one soft spot was camera/source coverage at 70.05%, tracked as a caveat rather than a blocker.

Three experimental milestones were run against this sample:

Baseline (EXP-01)

A straightforward transfer-learned fine-tune at 800px with a 10-layer frozen backbone. It established a working model but exposed a “confusion triangle” among visually adjacent classes — Blunt_Weapon, Melee_Weapon, and Tool were routinely mistaken for one another, and Person recall lagged badly.

Metric	Value
Precision	0.6327
Recall	0.4847
mAP50	0.5026
mAP50-95	0.3150

An intermediate “official” run (EXP-FINAL-07) attempted to confirm this configuration but actually regressed on three of four metrics (mAP50-95 fell to 0.2973), because the ten planned hyperparameter ablations behind it had never actually been executed — the tuning framework was built but its master switch was left off. Root-cause analysis (Notebook 08) traced the regression to the frozen backbone, the 800px resolution, and an unbracketed learning rate.

IMPROVED_1 (EXP-IMPROVED-02)

Acting on that diagnosis, the team fully unfroze the backbone and raised input resolution from 800px to 960px to better resolve small, distant objects — at the cost of a smaller batch size (16 → 12) and roughly 35% longer training.

Metric	Value
Precision	0.7350
Recall	0.6077

mAP50	0.6555
mAP50-95	0.4491

Gains vs. baseline: +10.2 points precision, +12.3 points recall, +15.3 points mAP50, +13.4 points mAP50-95.

IMPROVED_2 / Stage 11 Re-engineered (EXP-IMPROVED-03)

With the resolution and unfreeze strategy validated, the team went further: the taxonomy was consolidated from 7 to 5 classes, merging the confusion-triangle members (Blunt/Melee/Tool) into a single Handheld_Weapon category, batch size was raised to 24, and mosaic augmentation was progressively disabled (1.0 → 0.5 → 0.0) in favor of CCTV-specific photometric augmentation (low-light, blur, noise). Class weights were re-engineered for the new 5-class schema.

Metric	Value
Precision	0.7740
Recall	0.5940
mAP50	0.6570
mAP50-95	0.4300

Compared with IMPROVED_1, Stage 11 pushed precision higher (+5.3%) but gave back a little recall (-2.3%) and mAP50-95 (-4.3%) — a deliberate precision/recall trade tied to collapsing the ambiguous classes and removing mosaic's boundary noise.

Sandbox verdict: the Stage 08 validation gate (mAP50-95 ≥ 0.3150) was PASSED with the IMPROVED_1 configuration at 0.4491, a +42.56% margin above threshold — clearing the project to proceed to full-dataset training.

Chapter 2: The Scaling Challenge — Engineering Trade-offs

Validating a configuration on 20,000 images is not the same as running it on the full production dataset. The full corpus — 130,763 images, 294,950 annotations, 7 classes — is roughly 6.5× larger than the sandbox sample, and its class imbalance is severe: Firearm alone accounts for 117,864 boxes against only 9,527 Blunt_Weapon boxes, a 7.7:1 to 12.4:1 imbalance depending on how it's measured.

Running the sandbox's winning recipe unmodified — 960px resolution, immediate full-backbone unfreezing — against 130K images was not viable. At that scale it would produce severe memory pressure, a drastic drop in images-processed-per-second, and an unacceptably long training run. Rather than simply accepting a weaker configuration, the team compensated architecturally:

Resolution rollback, compensated by attention

Input resolution was scaled back to 800px to keep batch sizes and iteration speed sustainable at production scale. To offset the resulting loss of spatial detail — and preserve sensitivity to small, distant handheld objects — the YOLO11s architecture was modified to inject CBAM (Convolutional Block Attention Module) blocks at the P3, P4, and P5 feature levels. This added only a 0.075% GFLOPs overhead while sharpening the network's spatial and channel-wise focus exactly where small-object detail was being sacrificed by the resolution rollback.

Addressing extreme imbalance with Focal Loss

At production scale, simple class weighting was no longer judged sufficient against a ~7.7× imbalance between the majority (Firearm) and minority (Person) classes. The team adopted an elementwise Focal Loss (gamma=2.0, alpha=0.25) combined with mathematically audited inverse-frequency class weights, to concentrate learning signal on rare-class misses without letting easy background frames dominate the gradient.

A 10-epoch freeze warm-up

Because the CBAM layers are newly initialized (not COCO-pretrained), immediately unfreezing the entire network — as the sandbox recipe did — risked gradient shocks that would corrupt the valuable pretrained backbone weights. A 10-epoch freeze was introduced so the new CBAM layers and the modified detection head could adapt to the visual domain before the full network was released for end-to-end fine-tuning.

In short: 800px + CBAM + Focal Loss + freeze warm-up is not a retreat from the sandbox's findings — it is the production-scale translation of the same underlying goal (small-object and rare-class sensitivity), re-engineered to fit a 6.5×-larger, more imbalanced dataset within realistic compute constraints.

Chapter 3: Production Reality — The 130K Full Run

The production model — the CBAM-enhanced YOLO11s described above — is currently training against the full 130,763-image dataset (105,471 images in the training split), using an AdamW optimizer, cosine learning-rate scheduling, and a deterministic 80-epoch pipeline. This run is actively in progress, not yet complete.

The most recent telemetry shows the run paused after Epoch 11 (approaching Epoch 12), and the built-in engineering safeguards are behaving exactly as designed:

- At Epoch 10 (the final epoch under the frozen backbone), the model reached its best metrics for the frozen-backbone phase: Precision 0.52505, mAP50 0.41472.
- At Epoch 11, the scheduled unfreeze callback fired, releasing the full backbone for end-to-end training. As expected for this kind of transition, metrics temporarily collapsed (mAP50 dropping to 0.01327) as the optimizer began re-adapting the entire feature-extraction pipeline — including the newly warmed-up CBAM layers — to the CCTV domain.

This dip is a known, designed-for artifact of the freeze→unfreeze transition, not a failure signal — it is the same phenomenon the sandbox phase would have hit if it had gone straight to full unfreezing without the warm-up.

Production dataset profile

Metric	Value
Total images	130,763
Total annotations	294,950
Classes	7
Resolution	800 × 800
Firearm (majority class)	117,864 boxes
Blunt_Weapon (minority class)	9,527 boxes

Chapter 4: The Road Ahead

- Resume and complete training. 69 epochs remain in the deterministic 80-epoch schedule. The post-unfreeze metric collapse is expected to recover over the following epochs as the unfrozen backbone re-converges on the CCTV domain.
- Full evaluation suite. Once training concludes, generate the definitive confusion matrix, precision-recall curves, and per-class breakdowns on the full held-out set — the sandbox-phase confusion triangle (Blunt/Melee/Tool) and the Person class's weak recall are the specific risks to re-check at production scale.
- Export for deployment. Convert the final weights to an optimized inference format (e.g., TensorRT) suitable for real-time, edge-deployed CCTV inference.
- Outstanding items carried from the sandbox phase: an annotation-guideline audit for the Blunt/Melee/Tool confusion triangle, targeted data collection for the Person class, and formal isolation of each architectural change's individual contribution (CBAM, Focal Loss, freeze warm-up) now that they have been bundled into a single production run.

Sources: Weapon Detection CCTV System — YOLO11 Pipeline, Stages 03–09 (Notebooks 03–09 comprehensive report); Weapons-130K dataset summary; YOLO11s Experiment Evolution & Hyperparameter Matrix (sandbox visual comparison); normalized confusion matrix; Project Engineering Journey timeline.